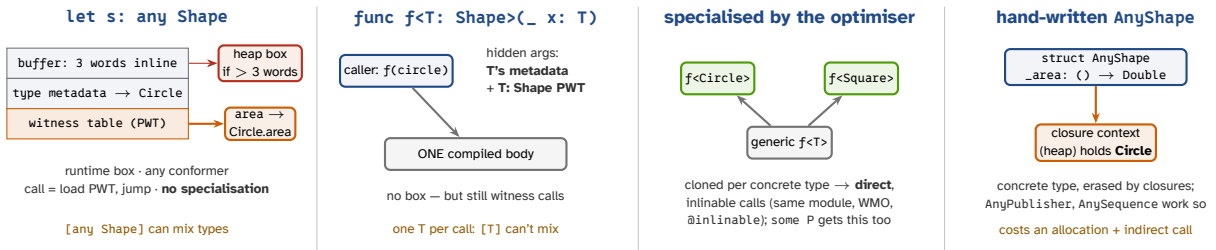


**In one line:** Swift turns runtime patterns into **compile-time** ones: a **phantom type/tagged ID** makes a wrong argument a compile error, an **enum** makes an illegal state unrepresentable, a **generic constraint** is a Strategy chosen at build time — and **type erasure** is the price of going back to runtime. Order of preference: **some/generics** > **any P<X>** > **a hand-written AnyX**.



**Type erasure — three tools** **Hand box:** store the conformer's methods as closures → a *concrete* type you can extend and conform (AnyPublisher, AnySequence). **any P<X>** (primary associated types, 5.7): a built-in box. **some P:** not erasure — hidden from the reader, known to the compiler. `eraseToAnyPublisher()` hides Publishers.Map&lt...>; AnyView also erases **structural identity** (worse diffing, lost @State); AnyHashable = mixed keys in one Set. An any P passed to <T: P> is **opened** (SE-0352).

```
protocol Loader<Output> { associatedtype Output
    func load() async throws -> Output }
struct AnyLoader<Output>: Loader { // hand-written box
    private let _load: () async throws -> Output
    init<L: Loader<_ base: L> where L.Output == Output {
        _load = base.load } // captures base
    func load() async throws -> Output { try await _load() } }
let b: any Loader<User> = RemoteUserLoader() // built-in (5.7)
```

**Phantom types · tagged IDs · newtype** A phantom parameter is never stored; it only stops mixing values with the same representation: **tagged IDs** (pointfree's Tagged<Tag, Raw>), **type-state** (a Request<Draft> has no send()). **Newtype:** struct Email with a failable init validates *once* — “parse, don't validate”. Zero runtime cost.

```
struct ID<Entity>: Hashable, Sendable { let raw: UUID }
struct User { let id: ID<User> }
struct Order { let id: ID<Order>; let buyer: ID<User> }
func user(_ id: ID<User>) -> User? { nil }
// user(order.id) error: ID<Order> is not ID<User>
enum Draft {}; enum Signed {} // uninhabited tags
struct Request<State> { var body: Data }
extension Request where State == Draft {
    func signed() -> Request<Signed> { .init(body: body) } }
extension Request where State == Signed { func send() async {} }
```

**Protocol witnesses — a struct of closures** One struct whose *properties* are *closures* replaces protocol + conformers; each “conformance” is a static value (.live, .mock). + override *one* endpoint in a test, derive variants by transforming values, no any/associated-type friction. — no default methods, no Self; closures must be @Sendable to share. pointfree's **swift-dependencies** clients are built this way.

```
struct TimeClient: Sendable {
    var now: @Sendable () -> Date
    var sleep: @Sendable (Duration) async throws -> Void }
extension TimeClient { static let live = TimeClient(
    now: { Date() }, sleep: { try await Task.sleep(for: $0) }) }
var t = TimeClient.live; t.now = { .distantPast } // one endpoint
```

**Builder: result builder vs fluent API** @resultBuilder (SE-0289): the compiler rewrites the closure's statements into buildExpression/buildBlock/buildOptional/ buildEither/buildArray calls — a compile-checked DSL (@ViewBuilder). **Fluent:** each call returns a modified copy of a struct — `Req().header(k, v).timeout(5); no build(), never half-built.`

```
@resultBuilder enum PathBuilder {
    static func buildExpression(_ s: String) -> [String] { [s] }
    static func buildBlock(_ c: [String]...) -> [String] {
        c.flatMap { $0 } }
    static func buildOptional(_ c: [String]?) -> [String] { c ?? [] }
}
func path(@PathBuilder _ make: () -> [String]) -> String {
    make().joined(separator: "/") }
let p = path { "users"; if isAdmin { "admin" } } // users/admin
```

**Closures as Strategy · enums as state machines** A one-method Strategy is a function: sorted(by:), a price closure. Use a protocol once there are *several* related operations, state, or it needs a name. **State machine:** each case carries only the data valid in that state (no isLoading + error? + items combinations); transitions = one switch on (state, event).

```
enum Load { case idle, loading, loaded([Item]), failed(String) }
enum Event { case start, success([Item]), failure(String) }
extension Load { mutating func send(_ e: Event) {
    switch (self, e) {
    case (.idle, .start), (.failed, .start): self = .loading
    case (.loading, .success(let v)): self = .loaded(v)
    case (.loading, .failure(let m)): self = .failed(m)
    default: break } } } // illegal transition: assert/log here
```

**Default implementations — the subclass trap** Extensions give defaults and mixins. If a **superclass** conforms *using* the default, the witness is bound for Base; Sub's method (no override needed!) is never reached through the protocol. **Fix:** implement it in Base; Sub must then write override.

```
protocol Greeter { func hi() -> String }
extension Greeter { func hi() -> String { "default" } }
class Base: Greeter {} // witness = the extension
class Sub: Base { func hi() -> String { "sub" } } // no override
let g: any Greeter = Sub()
g.hi() // "default" (but Sub().hi() is "sub")
```

**Generic constraint = compile-time Strategy** The strategy is a type fixed at the use site: specialised and inlined, no stored existential, no runtime swap.

```
protocol Backoff { static func delay(_ n: Int) -> Duration }
enum Exponential: Backoff { static func delay(_ n: Int)
    -> Duration { .milliseconds(100 << n) } }
struct Retrier<B: Backoff> {
    func wait(attempt n: Int) async throws {
        try await Task.sleep(for: B.delay(n)) } } // static call
let r = Retrier<Exponential>() // fixed at build time
```

## Traps

- ! AnyView everywhere: lost identity → worse diffing, reset @State.
- ! “some is type erasure” — no: the compiler still knows the type.
- ! Bare P still means any P; any is required for associated-type/Self protocols, everywhere with ExistentialAny.

## Remember

**Make wrong code not compile:** phantom → wrong ID · enum → wrong state · newtype → unvalidated input. **some** > **any** > **AnyX**.